Kolluru Suraj

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.15
BHASHYAM JUNIOR COLLEGE	Class XII (Board of intermediate education, Andhra Pradesh)	2024	90.90%
BHASHYAM HIGH SCHOOL	Class X (Board of secondary education, Andhra Pradesh)	2022	93.93%

SKILLS

Languages: C| mySQL | Verilog | Python |

Essential Technical Skills: Git | Linux | Latex

Tools used : MATLAB I LTspice | Intel Quartus | Cadence |VS code | Eagle | Ngspice | Altium

RELEVANT COURSES

Core Courses: Analog electronics, Digital Systems, Circuits and network analysis, Signal Processing, Semiconductor devices and fundamentals, Control systems, Engineering Electromagnetics Electrical machines, Power system Analysis.

Computer Science Courses: Introduction to programming, DSA, DBMS-1, Scientific Programming for Electrical Engineers.

Math Courses: Machine Learning, Mathematics for Electrical Engineers, Vector calculus, Probability & random processes, Matrix theory, Differential equations and Transform Techniques.

Lab Courses: Electrical Circuits Lab, Digital Systems Lab, Electronic Devices and Circuits Lab, DSP Lab, PCB Design and Layout for electronic systems

PROJECTS

Lock-In Amplifier

- Designed a **lock-in amplifier** using **AD630-based synchronous detection** for recovering weak AC signals buried in noise.
- Implemented **quadrature demodulation (I/Q)** for simultaneous extraction of amplitude and phase information.
- Developed a low-noise analog signal chain with variable-gain preamplification and active low-pass filtering stages.
- Designed a **frequency divider circuit** to generate precise 0° and 90° reference signals for synchronous detection.
- Simulated and validated the complete analog signal chain, achieving strong rejection of off-frequency interference.
- Designed the PCB in **Altium Designer.**

FIR Filter and FFT Processor on FPGA

- Implemented pipelined and non-pipelined architectures of a 100-tap fixed-point FIR filter in Verilog, optimizing the datapath and comparing resource utilization and maximum operating frequency.
- Designed and verified a pipelined 8-point Radix-2 FFT processor in Verilog, validating results against MATLAB simulations.
- Synthesized and deployed FIR and FFT designs on FPGA using Quartus Prime, performing timing analysis and comparing achievable clock frequencies across architectures.

Sampling and Reconstruction Demonstration using SAR ADC

- Designed a 4-bit SAR ADC with SAR logic, comparator, and R-2R ladder DAC to demonstrate analog-to-digital

conversion principles.
- Implemented and verified the successive-approximation conversion process using hardware and simulation-based testing.
- Demonstrated signal sampling, quantization, and analog reconstruction using DAC output and low-pass filtering, validating the complete sampling and reconstruction chain.

Analog Circuit Design and Simulation(LTSpice)

- Designed and implemented practical op-amp integrators, differentiators, Schmitt trigger, and waveform generators.
- Designed and implemented RC phase-shift and Wien bridge oscillators, analyzing their oscillation characteristics.
- Designed and implemented second-order Sallen-Key low-pass, high-pass, and band-pass filters, validating frequency responses.
- Designed and implemented a common-source amplifier, evaluating its DC biasing and small-signal performance.

POSITIONS OF RESPONSIBILITY

Class Representative (CR), EE24 Batch: Representing the EE24 batch as Class Representative, coordinating between students and faculty, streamlining academic communication, and assisting in the organization of departmental initiatives and student activities.

SPIC MACAY Multimedia Coordinator: Managed multimedia content, publicity campaigns, and digital outreach for SPIC MACAY IIT Hyderabad, contributing to increased visibility and engagement for cultural and heritage events.

MILAN Creatives Coordinator: Led the design of posters and promotional creatives for MILAN, IIT Hyderabad's annual General Championship. Supported event outreach and branding initiatives through digital content creation and publicity campaigns.

EXTRACURRICULAR

- Currently serving as a volunteer in the National Service Scheme (NSS), contributing to community service and social outreach initiatives.
- Hobbies include Gaming, cricket, and badminton.